

CHANGJIN, Korea, Democratic People's Republic of

WMO: 470310

Lat: 40.367N

Lon: 127.250E

Elev: 3543

StdP: 12.91

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.8	-12.1	-20.7	2.0	-14.3	-17.1	2.5	-10.4	21.1	3.1	19.9	4.3	2.3	0	0.558

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.2	78.7	67.5	76.3	66.4	73.8	65.1	70.6	75.8	69.0	73.7	67.4	71.7	5.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.0	121.7	73.9	67.4	114.8	71.7	65.9	109.0	70.2	37.0	76.0	35.5	74.0	34.1	72.1	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.6	17.1	14.3	-23.1	83.4	4.4	2.6	-26.3	85.2	-28.9	86.7	-31.3	88.1	-34.6	90.0		
			-22.4	72.9	4.6	1.7	-25.7	74.1	-28.4	75.2	-31.0	76.1	-34.4	77.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.0	6.2	11.3	24.1	38.1	50.5	59.0	65.4	64.9	54.7	42.4	26.8	11.2	(6)
	DBStd	21.86	8.07	8.60	9.31	7.32	6.11	4.41	4.57	4.59	5.20	6.19	9.05	8.74	(7)
	HDD50	5831	1358	1083	803	359	68	1	0	0	17	243	695	1204	(8)
	HDD65	9968	1823	1503	1268	807	451	184	51	58	310	699	1145	1669	(9)
	CDD50	1466	0	0	0	3	83	271	479	463	158	9	0	0	(10)
	CDD65	126	0	0	0	0	1	4	65	55	1	0	0	0	(11)
	CDH74	510	0	0	0	0	30	63	247	166	4	0	0	0	(12)
	CDH80	45	0	0	0	0	3	2	25	15	0	0	0	0	(13)
(14) Wind	WSAvg	3.7	4.0	3.9	4.3	4.7	4.7	3.3	2.8	2.6	2.7	3.5	4.1	4.0	(14)
(15) Precipitation	PrecAvg	30.8	0.3	0.6	0.6	1.6	2.6	3.8	8.2	6.3	3.8	1.5	1.2	0.6	(15)
	PrecMax	45.8	1.0	1.9	2.3	3.5	5.8	9.2	21.2	18.8	10.0	4.6	4.2	2.4	(16)
	PrecMin	17.7	0.0	0.0	0.2	0.1	0.3	1.1	1.3	1.8	0.4	0.1	0.1	0.0	(17)
	PrecStd	7.8	0.2	0.6	0.4	0.9	1.2	2.0	4.3	4.0	2.7	0.9	0.9	0.5	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.3	39.7	53.9	68.2	77.5	79.7	83.4	81.9	73.6	65.9	52.4	36.3	(19)
		MCWB	30.3	34.4	42.3	50.5	57.7	62.0	70.7	70.3	61.5	52.7	43.5	32.7	(20)
	2%	DB	27.8	34.1	47.8	61.6	72.4	75.5	79.2	77.9	69.8	60.7	46.9	32.5	(21)
		MCWB	24.5	30.2	38.8	48.1	54.8	60.9	69.2	68.8	59.3	50.2	40.3	30.1	(22)
	5%	DB	24.2	30.4	42.5	57.0	68.6	72.3	76.7	75.4	67.4	57.0	43.0	28.8	(23)
		MCWB	21.4	26.3	35.4	45.5	53.8	60.7	68.3	67.8	58.4	47.9	38.7	26.5	(24)
	10%	DB	20.9	26.9	38.1	52.5	64.4	69.4	74.1	72.8	64.7	53.3	39.4	25.3	(25)
		MCWB	18.7	23.7	32.5	43.0	52.2	60.3	67.3	66.4	57.0	46.4	35.8	23.0	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.9	35.3	44.0	55.3	61.3	67.2	73.3	72.7	66.0	55.6	45.9	33.8	(27)
		MCDB	32.7	38.7	51.0	64.2	71.5	74.0	79.4	78.9	70.7	62.1	50.3	35.7	(28)
	2%	WB	24.8	30.7	39.4	49.6	58.1	64.6	71.2	70.4	62.4	52.1	41.8	30.5	(29)
		MCDB	27.4	34.3	45.8	58.8	68.6	70.9	76.4	75.6	66.5	57.6	44.9	32.4	(30)
	5%	WB	21.7	26.8	35.8	46.4	55.7	62.8	69.6	68.9	60.2	49.5	39.0	26.4	(31)
		MCDB	23.9	30.0	42.0	55.8	65.1	69.1	74.5	73.5	64.4	54.9	42.3	28.6	(32)
	10%	WB	18.6	23.8	33.0	43.6	53.2	61.1	68.1	67.5	58.3	47.4	36.1	23.2	(33)
		MCDB	20.8	26.8	37.5	51.5	61.7	66.9	72.7	71.6	63.0	52.4	38.9	25.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	23.3	24.0	20.7	21.2	23.0	18.9	14.2	14.5	18.9	20.8	17.4	20.3	(35)
		MCDBR	24.1	25.5	24.3	30.5	32.1	25.8	18.8	17.7	21.7	26.7	21.6	20.6	(36)
		MCWBR	21.0	21.2	15.8	18.8	16.4	13.8	10.1	9.7	12.1	16.1	15.4	18.1	(37)
	5% WB	MCDBR	23.4	24.1	22.6	27.9	26.8	19.8	15.9	15.1	17.9	23.0	18.5	19.8	(38)
		MCWBR	20.9	20.4	15.6	18.5	16.2	12.6	9.7	9.5	11.6	15.7	14.4	17.7	(39)
(40) Clear-Sky Solar Irradiance	taub		0.323	0.390	0.461	0.529	0.553	0.595	0.538	0.479	0.424	0.428	0.374	0.326	(40)
	taud		2.079	1.923	1.813	1.748	1.750	1.717	1.927	2.085	2.162	2.024	2.078	2.115	(41)
	Ebn at Noon		255	251	246	239	237	227	239	251	257	236	233	243	(42)
	Edh at Noon		36	49	61	70	71	74	59	49	43	44	36	32	(43)
(44) All-Sky Solar Radiation	RadAvg		822	1113	1407	1591	1741	1592	1397	1397	1344	1089	753	687	(44)
	RadStd		42	66	95	121	159	153	173	141	130	80	92	43	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page